

Student Management System

Java program — Inheritance (UnderGraduate & PostGraduate)

Program 2: Student Management (Inheritance)

Demonstrates inheritance with a base Student class and two subclasses: UnderGraduate and PostGraduate, including validation checks.

Source Code

```
class Student{
    int id;
    String name;
    String dept;
    Student(int id,String name,String dept){
        this.id=id;
        this.name=name;
        this.dept=dept;
    }
    void displayCommondetails(){
        if (id<0){
            System.out.println("Invalid ID");
        }
        else{
            System.out.println("Student ID:"+id);
        }
        System.out.println("Student Name:"+name);
        System.out.println("Department:"+dept);
    }
}

class UnderGraduate extends Student{
    int sem;
    double cgpa;
    UnderGraduate(int id,String name,String dept,int sem,double cgpa){
        super(id,name,dept);
        this.sem=sem;
        this.cgpa=cgpa;
    }
    void displayUnderGraduatedetails(){
        displayCommondetails();
        if (sem<1||sem>8){
            System.out.println("Invalid Semester");
        }
        else{
            System.out.println("Semester:"+sem);
        }
        if (cgpa<0||cgpa>10){
            System.out.println("Invalid CGPA");
        }
        else{
            System.out.println("CGPA:"+cgpa);
        }
    }
}

class PostGraduate extends Student{
    String spl;
    String rt;
    PostGraduate(int id,String name,String dept,String spl,String rt ){
        super(id,name,dept);
        this.spl=spl;
        this.rt=rt;
    }
    void displayPostGraduatedetails(){
        displayCommondetails();
        System.out.println("Specializatio:"+spl);
        System.out.println("Research Type:"+rt);
    }
}

public class Main{
    public static void main(String[]args){
```

```

UnderGraduate ug = new UnderGraduate(101,"Ravi","CSE",5,8.75);
System.out.println("Undergraduate Details:");
ug.displayUnderGraduatedetails();
System.out.println();

PostGraduate pg=new PostGraduate(201,"Anita","ECE","AI","Computer Vision");
System.out.println("Postgraduate Details:");
pg.displayPostGraduatedetails();
System.out.println();

UnderGraduate stu=new UnderGraduate(102,"Kiran","ISE",3,8.23);
System.out.println("Student Details:");
stu.displayUnderGraduatedetails();
System.out.println();

UnderGraduate sd=new UnderGraduate(0,""," ",7,9.10);
System.out.println("UG Specific Details:");
sd.displayUnderGraduatedetails();
System.out.println();

PostGraduate sdd=new PostGraduate(0,""," ","Data Science","NLP");
System.out.println("PG Specific Details");
sdd.displayPostGraduatedetails();
System.out.println();

Student id=new Student(-10,"Rahul","CSE");
id.displayCommondetails();
System.out.println();

UnderGraduate sem=new UnderGraduate(0,""," ",10,0);
sem.displayUnderGraduatedetails();
System.out.println();

UnderGraduate cgpa=new UnderGraduate(0,""," ", 5,11.5);
cgpa.displayUnderGraduatedetails();
System.out.println();

PostGraduate rt=new PostGraduate(0,""," ","","");
rt.displayPostGraduatedetails();
System.out.println();
    }
}

```

Output

Undergraduate Details:

Student ID:101

Student Name:Ravi

Department:CSE

Semester:5

CGPA:8.75

Postgraduate Details:

Student ID:201

Student Name:Anita

Department:ECE

Specializatio:AI

Research Type:Computer Vision

Student Details:

Student ID:102

Student Name:Kiran

Department:ISE

Semester:3

CGPA:8.23

UG Specific Details:

Student ID:0
Student Name:
Department:
Semester:7
CGPA:9.1

PG Specific Details

Student ID:0
Student Name:
Department:
Specializatio:Data Science
Research Type:NLP

Invalid ID
Student Name:Rahul
Department:CSE

Student ID:0
Student Name:
Department:
Invalid Semester
CGPA:0.0

Student ID:0
Student Name:
Department:
Semester:5
Invalid CGPA

Student ID:0
Student Name:
Department:
Specializatio:
Research Type: